Supplementary Information

An open-source, 3D-printed auger powder doser designed with generative artificial intelligence

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S1 Bill of materials

Table S1 lists the complete per-channel bill of materials for the single-channel module, plus system-shared items (one per platform regardless of channel count). Prices are single-unit USD as of 2026-Q2 and should be reconfirmed at order time; vendor links and mirrored vendor design files are maintained in `hardware/` in the repository.

Table S1: Bill of materials. Printed parts are excluded (about 350 g PLA per channel, <$10), as is the gravimetric-feedback instrument (A&D HR-100A analytical balance, 0.1 mg readability), shared laboratory equipment that most labs already own.

Part	Qty	Price (USD)	Source
Per channel			
NEMA-11 bipolar stepper, 0.67 A, 10 N·cm (11HS18-0674S)	1	14	StepperOnline
Pololu Tic T500 stepper controller	1	33	Pololu #3135
JF-0530B 5 V push–pull solenoid (tap actuator)	1	7.50	Adafruit #412
DRV8871 H-bridge breakout (solenoid drive)	1	7.50	Adafruit #3190
ERM coin vibration motor, 10 mm	1	1.95	Adafruit #1201
DRV2605L haptic driver breakout (I^2C)	1	7.95	Adafruit #2305
MG996R-class metal-gear servo (tilt)	1	12	generic
6805ZZ bearing (auger support)	1	3	generic
5 mm shaft coupler, fasteners (M3/M5), bulk capacitors	—	10	generic
System-shared (one per platform)			
Raspberry Pi Pico W	1	6	Raspberry Pi
MAX3232 RS-232 transceiver breakout (balance interface)	1	3	generic
Mean Well GST60A12-P1J 12 V/5 A supply + cord + pigtail	1	26	Digi-Key
Pololu D24V22F5 5 V buck regulator	1	19	Pololu #2858
Pololu 33 V shunt regulator (back-EMF clamp)	1	15	Pololu #3776
Breadboard/protoboard, headers, wiring	—	10	generic
Total, single-channel platform		**≈175**	

S2 Construction guide

A step-by-step illustrated build guide is maintained in the repository (`https://github.com/vertical-cloud-lab/powder-doser`) alongside the parametric CAD sources and STLs for every part. In outline:

1. **Print** the auger, gear pair, bracket, tap collar (two clamp halves), mounting plate, hinged baseplate, and servo sector in PLA (0.2 mm layers, 4 perimeters, 25% infill, no supports; print the auger vertically, helix up).

2. **Auger drive:** press the 6805ZZ bearing into the bracket, seat the auger, mesh the printed gear pair, and bolt the NEMA-11 stepper to the front face of the mounting plate.

3. **Tap collar:** clamp the two collar halves around the auger tube; mount the solenoid to the collar flange (2×M3) so the plunger strikes the anvil pad, and adhere the ERM motor to the opposite pad.

4. **Tilt stage:** join the mounting plate to the baseplate through the side hinges (printed pins), engage the servo sector, and confirm the exit nozzle lies on the hinge axis.

5. **Electronics:** wire the Pico W, Tic T500 (UART), DRV8871, DRV2605L (I^2C), MAX3232 (RS-232 to the HR-100A balance), and servo per the KiCad schematic in `hardware/test-module/`; flash the MicroPython firmware.

6. **Checkout:** run the firmware self-test (rotate, tap, vibrate, tilt, tare/read), then a per-powder calibration sweep.

S3 Exit-nozzle variants

Four printed exit-nozzle geometries were produced for dispense testing (Fig. S1); the dispensing campaign will report flow-initiation and dribble behaviour per nozzle and per powder.

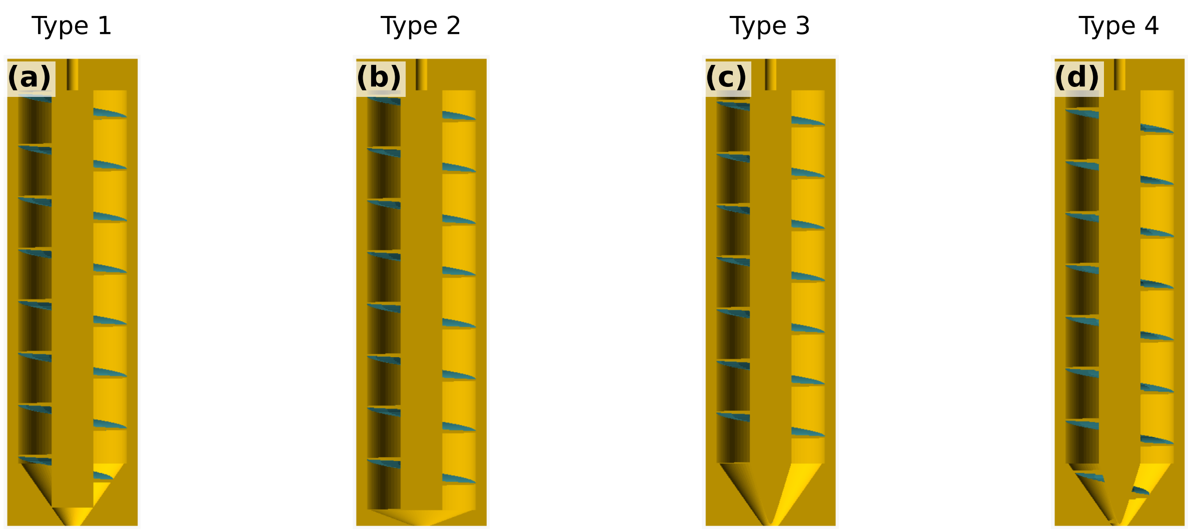

Figure S1: Cross-sections of the four exit-nozzle variant CAD models (Types 1–4, panels (a)–(d)) of the test auger, differing in exit constriction and chamfer geometry. The variants are parametric OpenSCAD models (`cad/auger-geared/nozzle-variants.scad`) authored by AI coding agents under human review; no GUI CAD package was used.

S4 AI usage log

In line with Digital Discovery's guidance on LLM use, all AI interactions are preserved verbatim and publicly:

- **CAD/code generation:** every agent session (GitHub Copilot coding agent; Claude-family models, April–June 2026) is recorded as a pull request containing the prompt (issue text and review comments), the generated CAD code, rendered views, and the human review thread.
- **Design reviews and literature queries:** Edison Scientific query scripts, task identifiers, raw answers, and extracted references are archived under `paper/background/` on the corresponding branches; the consolidated, validated bibliography with per-entry verification status is at `paper/references_validation.md`.
- **Text-to-CAD evaluations:** CADSmith run artifacts and the Zoo Design Studio (Zookeeper agent) session transcripts and best-practice notes are preserved on their respective branches.

- **Qualitative observations:** the running team journal of AI behaviour (good and bad) is public in the repository's discussion #39.